Data-Driven VMS-LES Modeling

An Overview of its History & Future

by

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Introduction

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The Navier-Stokes equations describe the motion of viscous fluid substances. Although they are universally applicable to fluids, a mathematical solution to the equations is yet to be derived, and they remain as one of the millennium prize problems. Due to its complex nature, engineers instead opt to model different parts of the equations, depending on the nature of the problem they wish to simulate. For Computational Fluid Dynamics (CFD), several approaches exist, ranging in physical fidelity and computation cost, from the simplest Reynolds-Averaged Navier-Stokes (RANS) to the Direct Numerical Simulation (DNS) of fluids. DNS scales rapidly with the Reynolds Number of the flow, computationally infeasible for most projects ...

Over the years, computational power increased exceptionally fast. However, as processors are starting to reach the atomic scale, it is evident that this rapid growth of computational power might come to a halt. It is due to this expected slowing down of computation power, new, innovative methods for simulating the NS equations are needed. For Large Eddy Simulation (LES), new approaches of coupling data-driving closure models to the numerical solver are being tested. Using projected DNS data, these data models might be able to learn the underlying physics, and become a universal model for subgrid-scale closure modeling. If successful, LES can become more practical and accurate, whilst being computationally cheaper than DNS. Ultimately, LES can be used to inform RANS in a similar fashion, improving RANS accuracy and allowing for more robust, yet relatively very cheap, simulations of fluids.

Coupling an Artificial Neural Network (ANN) to a Variational Multiscale (VMS) formulation has been attempted prior. First, Robijns [**Robijns2019**] laid the groundworks and established the fact that strong *a priori* correlations between predicted ANN closures and projected DNS closures can be achieved. However, later Janssens [**Janssens2019**] showed that while strong a priori correlations are possible, these are not directly translated to *a posteriori* performance. In fact, using the ANN predictions led to two critical instability mechanisms: Corrector-Pass Instability (CPI) and Long-Term Instability (LTI). Rajampeta [**Rajampeta2021**] was able to remove the CPI from the

Newton corrector passes by the use of a Lagged Feature Set (LFS), and used backscatter limiting to address the LTIs. He was successful in achieving a stable online run, but backscatter limiting is unphysical in its nature, motivating future work for discovering other methods which could address the LTI problem. Bettini [**Bettini2023**] attempted a more energy-constrained approach, which led to over dissipative simulation runs. Krochak [**Krochak2024**] continued the work, attempting three different stabilization strategies, yet the LTI still remained. Dominik...

cite dominik

From previous work, it is clear that the problem of LTIs is detrimental, yet persistent. For this work, a new approach of including an additional artificial dissipation term is taken. This artificial term is intended to correct the erroneous predictions of the model, and to drain the accumulating energy, in order to hopefully bring balance to the force.

Check the goal of this artificial dissipation term, is this how we are gonna use it for?

Continuing the persistent problem of solver instability when using VMS-ANN coupled models, the research proposal is defined as follows.

Research Proposal

Research Plan

In order to further specify the proposal, the following research questions are formulated:

1. **Question 1**
   - 1.a Subquestion 1.1
   - 1.b ...
2. **Question 2**
3. **...**

2

Problem Description

3

Literature Review

3.1. Computational Fluid Dynamics

add citations where needed

CFD simulations are used to solve various fluid dynamic problems. An example of applying CFD simulations is the Navier-Stokes equations. These equations govern the flow of fluids over bodies (or without them) and are thus important in many engineering practices, from aeronautical to automotive. Unfortunately, no analytical solution exists, so engineers must approximate these equations numerically. The most accurate numerical method is DNS. Here, the solver resolves all flow scales, yielding the most accurate results but at a high computational cost. In fact, representing all scales increases computational costs very quickly. The smallest scales, also known as the Kolmogorov microscales, are all proportionate to the Reynolds number, Re. For three-dimensional DNS, costs are proportionate to Re^3. Thus, for large-scale flow problems, DNS is computationally infeasible, and it is currently used primarily as a research tool for studying turbulent phenomena.

To reduce computational costs, researchers have derived a method that splits the solution into a mean and a fluctuating part. This method, called RANS, is among the least expensive simulations in the CFD landscape. Here, the solver computes the mean solution while accounting for fluctuations through modeling. Despite various turbulence models, RANS simulations struggle to incorporate the influence of the fluctuating flow components on the mean, especially for highly dynamic and wall-bounded problems.

A third method of solving fluid dynamics problems is to use LES. This method lies between the low-fidelity, cheap RANS simulations and the high-fidelity, expensive DNS simulations. Similarly to DNS, the solver resolves the flow scales, but only the largest, most influential ones. LES is thus a DNS, with a spatial filter applied, which removes the smaller, and more expensive scales. This approach makes LES much cheaper than DNS, but it does remove some physical fidelity. Naturally,

if greater physical accuracy is required, the filter width can be reduced, allowing LES to approach and eventually become DNS.

This approach splits up the solution into the larger scales, also known as the resolved scales, and the smaller, Subgrid-Scale (SGS) terms. The solver models the influence of SGS terms. Modeling is necessary because smaller scales serve two important functions. One, they drain the flow field of energy through dissipative phenomena. Two, in specific scenarios, they re-transfer energy to the larger scales through a process known as *backscatter*. The standard modeling approach is to use the Smagorinsky model. This pioneering model is relatively simple and correctly captures energy cascades in simulations, making it a great option. However, the model has several drawbacks, including limited wall behavior, no backscatter, and a tendency to be overly dissipative. Other models exist, such as the Dynamic Smagorinsky and Scale Similarity models.

The problem with classical LES is that the solver applies the influence of the model to the complete flow domain, which tends to make simulations overly dissipative. A different approach, developed by Hughes, uses the variational multiscale framework.

3.1.1. Variational Multiscale Framework

add citations where needed

VMS is an unconventional approach to LES. Instead of applying a low-pass spatial filter, the approach performs a finite-element projection, which splits the continuous unknown into a resolvable component and a subgrid component. This projection naturally introduces the spatial filter. The framework has several advantages over a classical LES formulation. First, the effect of modeling is confined only to the smallest resolved scales, which can reduce the effect of modeling errors. Second, with a stronger mathematical foundation, it can perform better in wall-bounded geometries.

Consider a general nonlinear Partial Differential Equation (PDE) written in residual form:

$$\mathcal{R}(u) = 0 \tag{3.1}$$

where $u \in \mathcal{V}$ is the solution in a suitable function space (e.g. Sobolev space) and $\mathcal{R}(u)$ is the residual operator. The variational formulation reads:

Find $u \in \mathcal{V}$ such that

$$(\mathcal{R}(u), w) = 0 \quad \forall w \in \mathcal{V} \tag{3.2}$$

where w are test functions and $(.,.)$ denotes an inner product over the domain. VMS decomposes the solution space into

$$\mathcal{V} = \bar{\mathcal{V}} \oplus \mathcal{V}' \tag{3.3}$$

Where $\bar{\mathcal{V}}$ are the resolved (coarse) scales and $\mathcal{V}'$ are the unresolved (fine/subgrid) scales. Correspondingly,

$$u = \bar{u} + u' \qquad w = \bar{w} + w' \tag{3.4}$$

Substituting this decomposition back into the weak form yields two coupled variational equations: a coarse-scale equation and a fine-scale equation. The fine-scale equation can be substituted into the coarse-scale equation, yielding the final form. Solvers must approximate the contribution from the fine-scale equation.

3.2. Artificial Intelligence

add citations where needed

Artificial Intelligence (AI) is the simulation of human intelligence by computer systems to perform tasks like learning, reasoning, problem-solving, and decision-making, and is currently rapidly evolving due to recent breakthroughs. Applications of AI are extensive and diverse, ranging from interpreting patterns in speech to analyzing medical X-rays to detect anomalies, etc. Using Machine Learning (ML), they train on vast amounts of data. ML enables systems to automatically learn, analyze patterns, and improve from data without being explicitly programmed. This section explains ANNs plainly, as well as giving a basic overview of ML and the Multi-Layer Perceptron (MLP).

3.2.1. Artificial Neural Networks

add citations where needed

ANNs are humanity's best attempt at creating a digital analogy to the human brain. As the human brain is capable of performing complex tasks, such as recognizing faces in an instant, it is no wonder that humans are motivated to create models that mimic its behavior, and hopefully, its performance. Describing the human nervous system can, in essence, be done as a three-block system: receptors, effectors, and, centrally, the neural net, which is composed of neurons. Although neurons are much slower than silicon logic gates, the brain can compensate by having an enormous number of densely interconnected neurons. It is this dense interconnected neural network that inspired the mathematical models used to create ANNs.

Inspired by the human brain, ANNs can be described as systems of parallel, distributed proces-

sors, each composed of simple processing units. It acquires knowledge through a learning process in which it interacts with its environment and adapts the strengths of its interneuron connections (synaptic weights). Many types of ANN models exist using this definition. ANN is thus an umbrella term encompassing various models, each with its own advantages and disadvantages.

One of the simplest types of models is the feed-forward neural network, also called the MLP. Other than MLPs, well-known types of models are Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) models. CNNs are used for tasks such as predicting data types, including text, images, and audio. By applying filters, they create feature maps that can detect patterns such as edges, lines, and textures. RNNs on the other hand, use internal loops to retain memory of previous inputs. This approach makes them better suited for language translation and speech recognition. Of course, many other types of models exist, as the AI field is currently rapidly evolving.

As AI models are statistical approximators, they require large amounts of data for training. Not only that, but data has to be split up into three components: the actual training set, a validation set, and a testing set. The actual training set is indeed the dataset the model uses for learning and parameter updates. Next, the validation set decides when training should halt. It does not influence the weights; instead, it evaluates the model's current performance during training. The last dataset, the test set, is used once after training is complete. It evaluates the model's true generalization performance. The model must never see this set during training or tuning.

Indeed, the goal of the model is generalization, meaning it can perform well on similar datasets not seen during training. The opposite of generalization is overfitting. When a model is overfit, it only performs well on the training dataset. The model may have learned the underlying relationship, but it may also have learned the noise in the training dataset. It has memorized the training samples instead of capturing the true structure. Causes of overfitting can include too little data or too many training epochs. Common solutions include early stopping, regularization, and dropout.

3.2.2. Multilayer Perceptron Models

add citations where needed

MLP, also known as feed-forward neural networks, are one of the simplest frameworks in the ANN landscape. They consist of three main layers: the input layer, the output layer, and the hidden layer(s) in between. Each of these layers contains an arbitrary number of neurons, and each neuron in one layer connects with every node in the following layer. Their connections are mathematically presented by

$$z = w^T x + b \tag{3.5}$$

$$a = \sigma(z) \tag{3.6}$$

where x is the input vector, w are the weights, b is a bias term, and z represents the linear combination of the input, weights, and bias. z is then passed on to an activation function, σ, which is typically a sigmoid, tanh, or ReLU. a is the final neuron output, which becomes the input for the next layer of neurons. The input signal thus propagates forward through the model, passing each hidden layer until reaching the output layer.

$$x \to a^{(1)} \to a^{(2)} \to \ldots \to \hat{y}$$

During training, comparing the predicted value $\hat{y}$ to the true value y provides an error signal that propagates backward through the model. During backpropagation, the error contributed to each weight is computed, providing a measure of how to adjust it. One forward-backward loop is called an epoch. After many epochs, training stops when the stopping criteria are met.

As they are the simplest types of models, they will be used as the baseline model for predicting the SGS closure terms, as explained in Chapter **??**.

3.3. Machine Learning

add citations where needed

ANNs are trained on huge amounts of data to approximate underlying functions, recognizing data structures, etc. These models need to be informed of the accuracy of their predictions, providing the signals required for backpropagation. There are mainly three learning paradigms: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning requires a teacher to provide the models with correct target responses. On the contrary, unsupervised learning has no teacher. Instead, models train to discover and classify data structures. Lastly, reinforcement learning does not have a teacher present during learning; however, it differs from unsupervised learning in that the model still has to predict a continuous value. Reinforcement learning is thus a distinct learning philosophy, making it incorrect to classify it as supervised or unsupervised learning.

3.3.1. Supervised & Unsupervised Learning

add citations where needed

Supervised learning is currently the most studied machine learning paradigm, thanks to its straightforward nature. The main distinction from other paradigms is the presence of a teacher.

In other words, the availability of labeled training examples. Each example consists of a unique input signal and a corresponding desired response (target). The network's parameters are adjusted to minimize the error signal between the input and the target. This learning paradigm typically applies to regression problems, where the model predicts continuous values from input data.

Unsupervised learning is an approach in which a model learns patterns from unlabeled data. Typical objectives include grouping similar data, reducing dimensionality, or finding underlying features or structures. These objectives are called classification problems.

3.3.2. Reinforcement Learning

add citations where needed

Reinforcement Learning (RL) consists of an interaction loop between an *agent* and an *environment*. The agent decides what actions to take, based on its current state, which it receives from the environment. The environment can be more abstract than just the outside world. For example, say an agent is a robot with a battery. Even though the battery is internal to the robot, the charge status should still be considered part of the environment by the agent and influence the agent's decisions. Besides the agent and the environment, RL consists of four main sub-elements: a policy, a reward signal, a value function, and, optionally, a model of the environment. A policy defines the actions the agent will undertake given its current state and, roughly speaking, can be seen as a mapping from the agent's current state to its actions. A reward signal is the instantaneous goal of a learning problem. Usually, it is a scalar signal received at every step the agent takes. Agents will need to balance instantaneous rewards with long-term returns. The value function measures how beneficial it is for an agent to be in a particular state, taking into account the future rewards it can still obtain. Finally, the model of the environment describes how the environment behaves when the agent takes an action. It predicts the next state and the associated reward. Reinforcement learning splits into two methods: model-based and model-free, depending on whether a model is available. Of the two, model-free is more relevant to the problem defined in Chapter **??**.

Reinforcement learning contains several sub-methods for calculating rewards, updating policies, etc. Ranging from simpler tabular methods such as Dynamic Programming and Monte Carlo, to more complex methods, such as Implicit Q-Learning (IQL), Conservative Q-Learning (CQL), and TD3+BC (Behavior Cloning Regularization). For the research proposal, the latter methods are more relevant as the state spaces is tremendous, and offline off-policy learning is expected to yield the best performance while remaining computationally feasible.

Soft Actor-Critic

Proximal Policy Optimization

A3C / A2C

Dreamer / Model-based RL

AWAC

Advanced Offline Reinforcement Learning Algorithms

IQL addresses the distribution shift problem in offline reinforcement learning by avoiding the maximization over out-of-distribution actions that appears in standard Q-learning. Instead of computing $\max_a Q(s, a)$, IQL learns a state-value function $V(s)$ directly from the dataset using *expectile regression*. This allows the algorithm to estimate the value of high-quality dataset actions without explicitly evaluating unseen actions. The Q-function is then trained using the target $r + \gamma V(s')$, eliminating the need to query the Q-network for actions outside the dataset support.

Policy improvement is performed using advantage-weighted behavior cloning. Actions with higher estimated advantage are imitated more strongly, while low-advantage actions are suppressed. As a result, policy improvement occurs strictly within the dataset distribution. This greatly improves training stability compared to standard actor–critic methods and avoids the need for explicit conservative penalties. IQL has been shown to perform competitively on a range of offline RL benchmarks, particularly when datasets contain moderately good trajectories or demonstrations [**Kostrikov2021**].

cite this algorithm

Conservative Q-Learning CQL modifies the standard Q-learning objective to learn Q-values that are conservative with respect to actions not supported by the dataset. Offline RL methods often suffer from severe overestimation of Q-values because neural networks extrapolate poorly to Out-of-Distribution (OOD) actions. When these overestimated values are used during policy improvement, the resulting policy may exploit unrealistic actions that never occur in the dataset.

CQL addresses this problem by introducing a regularization term that explicitly pushes down the Q-values of unseen actions while maintaining high Q-values for dataset actions. This effectively ensures that the expected value of a policy under the learned Q-function lower-bounds its true value. In practice, the conservative penalty reduces the likelihood that the learned policy selects actions outside the dataset support.

Because of this pessimistic value estimation, CQL is particularly robust when the dataset is noisy, narrow, or partially random. Empirical results demonstrate that CQL consistently outperforms several earlier offline RL algorithms, including Soft Actor–Critic (SAC), BEAR, BRAC, and behavior cloning baselines on a range of benchmark tasks [**kumar2020**]. The trade-off is that the conservative regularization increases computational cost and may introduce some degree of underestimation bias.

cite these algorithms

Twin Delayed Deep Deterministic Policy Gradient (TD3) was originally introduced to address

overestimation bias in deterministic actor–critic algorithms. The method uses clipped double Q-learning to reduce optimistic value estimates. Instead of relying on a single critic, TD3 maintains two Q-functions and uses the minimum of the two when constructing the Bellman target:

$$y = r + \gamma \min_{i=1,2} Q_{\theta'_i}\big(s', \pi_{\phi'}(s')\big) \tag{3.7}$$

Using the minimum of the critics prevents the value target from introducing additional overestimation. Although this may introduce a slight underestimation bias, such bias is generally preferable because underestimated actions are less likely to be amplified through policy updates. TD3 further improves stability through the use of delayed policy updates, target networks, and target policy smoothing regularization, which reduce variance and prevent the policy from exploiting narrow peaks in the learned Q-function [**Fujimoto2018**].

To adapt TD3 to the offline setting, Fujimoto and Gu (2021) introduced Twin Delayed Deep Deterministic Policy Gradient with Behavior Cloning (TD3+BC), which augments the policy objective with a behavior cloning regularization term. The policy is trained to both maximize the Q-value and remain close to actions observed in the dataset:

$$\pi = \arg\max_{\pi} \mathbb{E}_{(s,a)\sim\mathcal{D}} \left[\lambda Q(s, \pi(s)) - \|\pi(s) - a\|^2\right] \tag{3.8}$$

The hyperparameter λ controls the strength of the behavior cloning constraint. This simple modification significantly improves stability in offline settings by preventing the policy from deviating too far from the dataset distribution. Despite its simplicity, TD3+BC performs competitively with more complex offline RL algorithms on several benchmark tasks while being substantially easier to implement and computationally cheaper [**Fujimoto2021**].

Table 3.1: Comparison of CQL, IQL, and TD3+BC for offline reinforcement learning.

Property	**CQL**	**IQL**	**TD3+BC**
Handles poor datasets	⋆⋆⋆	⋆⋆	⋆
Training stability	⋆⋆	⋆⋆⋆	⋆⋆
Computational cost	High	Medium	Low
Implementation complexity	Medium	Medium	Very Low
Policy improvement ability	Medium	Medium	High
Extrapolation robustness	⋆⋆⋆	⋆⋆	⋆

3.4. CFD-ANN Enriched Models

3.4.1. Other work

3.4.2. Past Work at TU Delft

The problem of solver stability has been persistent. Although multiple authors have contributed to the stability issue, the problem of long-term instability remains. In this section, the previous work of the authors at the TU Delft is explained, laying the groundwork for future progress and identifying the core issues that cause instability.

The first to try coupling a MLP with a LES solver to learn the unresolved-scale closure terms within a VMS formulation was Robijns [**Robijns2019**]. He attempted an unconstrained approach, tasking a ANN with representing the complete unresolved-scale model. In contrast, a constrained approach has been attempted prior to Robijns (Durieux, Beekman, Kurian). Of course, hybrid models are also conceivable. The unconstrained approach will remain the basis for future work. Using the one-dimensional Burgers equation as a substitute for the Navier-Stokes equations, Robijns demonstrated that neural networks are capable of learning the underlying physical phenomena, and achieved strong *a priori* correlations between ANN predictions and DNS projected closures. However, he performed no rigorous *a posteriori*, which led to subsequent work by the other authors.

Inserting the ANN predicted closures into an *online* simulation revealed severe stability issues. Janssens [**Janssens2019**] set out to investigate the sources responsible for this critical stability problem. He identified two fundamental instability mechanisms: CPI, caused by ANN-induced spurious roots in the Newton loop, and LTI, caused by error accumulation. Janssens essentially shifted the research focus from "can a neural network learn the underlying physics?" to "can it remain stable in an online setting?". From this point on, it was clear that strong *a priori* correlations do not translate directly to *a posteriori* performance.

find more sources for noise-added machine learning training

A tactic to increase robustness and stability, and reduce overfitting, of an ANN is to inject the training data with artificial noise. For the VMS-ANN coupled problem, the idea was to train the neural network on "perfect" projected DNS data. This training method prevented the model from ever encountering erroneous data, leaving it unprepared for situations where the input data contains self-induced errors. Pusuluri [**Pusuluri2020**] attempted this approach and found that injecting the training data with Gaussian noise indeed improved robustness and reduced sensitivity to off-manifold states encountered during online simulations. However, this came at a cost of model accuracy, and ultimately, the fundamental problem of LTI persisted.

Next, Rajampeta [**Rajampeta2021**] focused on fixing the CPI problem. He introduced the LFS strategy to decouple the ANN prediction from the corrector loop. This approach addressed the CPI problem, but LTI remained, for which he introduced backscatter limiting to prevent destabilizing energy transfer from the fine to the coarse scales. Limiting enabled stable simulations but reduced physical fidelity, as backscatter is inherent.

Bettini [**Bettini2023**] was the first to attempt the ANN-VMS formulation to a three-dimensional

turbulent channel flow. His idea for addressing the LTIs was to introduce an energy-constrained training formulation. Again, Bettini achieved strong *a priori* correlations, but *a posteriori* simulations showed excessive dissipation and persistent stability issues. Indeed, energy-consistent training alone is not sufficient to guarantee physically correct long-term behavior.

Krochak [**Krochak2024**] intended to redefine the 3D formulation, focusing on stability, solver consistency, and computational practicality. By reformulating the closure definitions, he demonstrated that modeling the continuity closure terms is particularly problematic. He evaluated three stabilization strategies: filtering of the ANN target, mini-batch sum loss, and rotational data augmentation. These improved robustness, but again, the problem of long-term instability remained.

Include Dominik in timeline

Table 3.2: Overview of past TU Delft work.

Author	Year	Problem	Approach
Robijns	2019	1D Burgers	VMM-ANN coupling feasibility
Janssens	2019	1D Burgers	*Online* analysis
Pusuluri	2020	1D Burgers	Noise-augmented training
Rajampeta	2021	1D Burgers	LFS & limiting
Bettini	2023	TCF Re_{180}	Energy-constrained
Krochak	2024	TCF	Filtering, mini-batch loss, rotational augmentation
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